Last-Iterate Guarantees for Learning in Co-coercive Games

Siddharth Chandak[1†], Ramanan Tamizholi[2†] and Nicholas Bambos[3]

***Abstract*— We establish finite-time last-iterate guarantees for vanilla stochastic gradient descent in co-coercive games under noisy feedback. This is a broad class of games that is more general than strongly monotone games, allows for multiple Nash equilibria, and includes examples such as quadratic games with negative semidefinite interaction matrices and potential games with smooth concave potentials. Prior work in this setting has relied on relative noise models, where the noise vanishes as iterates approach equilibrium, an assumption that is often unrealistic in practice. We work instead under a substantially more general noise model in which the second moment of the noise is allowed to scale linearly with the squared norm of the iterates, an assumption natural in learning with unbounded action spaces. Under this model, we prove a last-iterate bound of order $O(\log(t)/t^{1/3})$, the first such finite-time bound for co-coercive games under non-vanishing noise. We additionally establish almost sure convergence of the iterates to the set of Nash equilibria and derive time-average convergence guarantees.**

I. Introduction

Decision-making in multi-agent systems arises in many applications, including communication networks, distributed control, and economic markets. In these settings, each agent's outcome depends on both its own action and the actions of others, making strategic interaction a central feature. Game theory provides a natural mathematical framework for analyzing these interactions, where each agent seeks to maximize a utility function that captures its individual objective. In particular, *continuous games*, where agents select actions from continuous spaces, are especially relevant for modeling real-valued decision-making problems such as resource allocation, pricing, and distributed learning.

An important question in such games is whether agents can learn a Nash equilibrium (NE) through repeated decentralized interaction. In continuous games, gradient-based methods provide a natural approach to decentralized learning due to their simplicity and scalability, as each agent updates its action using only local gradient information of its utility function. In many practical settings, however, this gradient feedback is noisy due to factors such as observational errors, random disturbances, or stochasticity in the underlying utility functions. The most natural learning rule in this setting is stochastic gradient descent (SGD), or equivalently stochastic gradient ascent in the context of utility maximization.

To establish convergence guarantees for learning dynamics, it is necessary to impose structural assumptions on the underlying game. Most existing convergence and finite-time analyses for stochastic gradient methods have focused on *strongly monotone games*. Strong monotonicity of the game gradient operator enables a significantly sharper convergence analysis, yielding an $\mathcal{O}(1/k)$ convergence rate for stochastic gradient methods, and moreover guarantees uniqueness of the Nash equilibrium. Despite the tractability of this setting, the strong structural assumptions and uniqueness of equilibrium considerably restrict its scope of applicability.

In this work, we consider *co-coercive games*, a broader class that has attracted recent interest in the literature. Co-coercivity is a substantially weaker condition on the game gradient operator than strong monotonicity, making convergence analysis for stochastic gradient methods considerably harder. Moreover, co-coercive games may admit non-singleton Nash equilibrium sets, introducing an additional challenge absent in the strongly monotone setting. This class includes several important examples, such as quadratic games with negative semidefinite interaction matrices and potential games with smooth concave potentials, which we present as motivating examples in the paper.

In this work, we establish the first finite-time last-iterate convergence guarantee for *co-coercive games* under stochastic gradient dynamics with non-vanishing noise. Specifically, for the vanilla SGD algorithm, we prove a last-iterate mean-square bound of order $\mathcal{O}\left(\log(t)/t^{1/3}\right)$. In contrast to prior works on co-coercive games, which rely on relative noise models where the noise vanishes as iterates approach equilibrium, our analysis is carried out under a general state-dependent noise model in which the second moment of the noise scales linearly with the squared norm of the current iterate, an assumption that is natural in stochastic optimization and learning with unbounded action spaces, and that strictly subsumes the relative noise model as a special case. We additionally establish almost sure convergence of the iterates to the set of Nash equilibria and derive time-average convergence guarantees. Our analysis draws on techniques from the study of inexact Krasnosel'skii-Mann iterations, extended with additional arguments to handle last-iterate convergence under the general noise model.

II. Co-coercive Games

Consider a set of players $\mathcal{N} = \{1, 2, \ldots, N\}$, where each player n chooses an action $x_n = (x_n^{(1)}, \ldots, x_n^{(d)}) \in \mathbb{R}^d$. We denote by $\mathbf{x} = (x_1, \ldots, x_N) \in \mathcal{X} = \mathbb{R}^{Nd}$ the concatenation of all action vectors. A continuous game is defined as follows.

Definition 1. A continuous game is a tuple $\mathcal{G} =$

† Equal contribution. Listed alphabetically.

[1,3] Department of Electrical Engineering, Stanford University, USA
[2] Indian Institute of Science, Bengaluru, India
[1]chandaks@stanford.edu
[2]ramanant@iisc.ac.in
[3]bambos@stanford.edu

$(\mathcal{N}, \mathcal{X}, \{u_n\}_{n=1}^N)$ where $\mathcal{N}$ is the set of N players, $\mathcal{X} = \mathbb{R}^{Nd}$ is the set of action profiles and $u_n : \mathcal{X} \mapsto \mathbb{R}$ is the n-th player's utility function.

The properties of the game are characterized through the gradient operator induced by the utility functions. We assume that the function $u_n(\mathbf{x})$ is continuously differentiable in $\mathbf{x}$ for each $n \in \mathcal{N}$. Our key assumption is that the game is co-coercive. For this, we define the individual gradient as $v_n(\mathbf{x}) = \nabla_{x_n} u_n(\mathbf{x})$, and the game gradient operator as

$$\mathbf{v}(\mathbf{x}) = (v_1(\mathbf{x}), \ldots, v_N(\mathbf{x})).$$

Definition 2. For $\lambda > 0$, a continuous game with continuously differentiable utility functions is called λ-co-coercive if the following holds for all $\mathbf{x}, \mathbf{x}' \in \mathcal{X}$

$$\langle \mathbf{x}' - \mathbf{x}, \mathbf{v}(\mathbf{x}') - \mathbf{v}(\mathbf{x}) \rangle \leq -\lambda \|\mathbf{v}(\mathbf{x}') - \mathbf{v}(\mathbf{x})\|^2.$$

This sign convention is natural for utility maximization. Equivalently, the operator $-\mathbf{v}$ is λ-co-coercive in the standard monotone operator sense [1]. Before making further remarks on co-coercive games, we present the definition of Nash equilibrium (NE) for completeness.

Definition 3. An action profile $\mathbf{x}^*$ is a Nash equilibrium for game $\mathcal{G}$ if for each player $n \in \mathcal{N}$, $u_n(x_n^*, \mathbf{x}^*{}_{-n}) \geq u_n(x_n, \mathbf{x}^*{}_{-n})$ for all $x_n \in \mathbb{R}^d$.

Unlike strongly monotone games, where there is a unique NE, there can exist multiple or no NE in a co-coercive game. For co-coercive games with unconstrained action spaces, any Nash equilibrium $\mathbf{x}^*$ satisfies the first-order condition $\mathbf{v}(\mathbf{x}^*) = 0$ [2]. We denote the set of Nash equilibria by $\mathcal{X}^* = \{\mathbf{x} \in \mathbb{R}^{Nd} \mid \mathbf{v}(\mathbf{x}) = 0\}$. As our objective in this paper is to study learning on NE in co-coercive games, we make the following assumption that the set of NE is non-empty.

Assumption 1. *The set of NE, $\mathcal{X}^*$, is non-empty.*

A. Examples of Co-coercive Games

In this subsection, we present two examples of co-coercive games which are not strongly monotone. These help us better understand co-coercive games and build some intuition.

1) Quadratic Games with NSD Interaction Matrix: Another important class of co-coercive games is given by quadratic games. For simplicity, we consider the scalar-action case, where each player chooses an action $x_n \in \mathbb{R}$. The same construction extends naturally to the case of multi-dimensional actions by replacing scalar coefficients with block matrices.

Let $Q = (Q_{nm}) \in \mathbb{R}^{N \times N}$ be a symmetric negative semidefinite matrix, and define the utility of player n as

$$u_n(\mathbf{x}) = \frac{1}{2} Q_{nn} x_n^2 + \sum_{m \neq n} Q_{nm} x_n x_m.$$

Then, the game gradient operator is linear and given by

$$\mathbf{v}(\mathbf{x}) = Q\mathbf{x}.$$

Then the game is $\big(1/|\lambda_{\min}(Q)|\big)$-co-coercive, where $\lambda_{\min}(Q)$ denotes the smallest eigenvalue of Q [1, Pg. 79].

An illustrative example is given by the following two-player non-separable game:

$$u_1(x_1, x_2) = -\frac{1}{2}x_1^2 - x_1 x_2, \quad u_2(x_1, x_2) = -\frac{1}{2}x_2^2 - x_1 x_2.$$

Here, the corresponding interaction matrix

$$Q = \begin{pmatrix} -1 & -1 \\ -1 & -1 \end{pmatrix}$$

is negative semidefinite with eigenvalues -2 and 0. Hence, the game is co-coercive but not strongly monotone. The set of NE for this game is given by

$$\mathcal{X}^* = \{(x_1, x_2) \in \mathbb{R}^2 : x_1 + x_2 = 0\},$$

which illustrates that co-coercive games may admit infinitely many NE.

2) Potential Games with Concave and Smooth Potential: An important class of co-coercive games is given by potential games with a smooth concave potential function. A continuous game is called a *potential game* if there exists a continuously differentiable function $\Phi : \mathcal{X} \to \mathbb{R}$ such that

$$v_n(\mathbf{x}) = \nabla_{x_n} \Phi(\mathbf{x}), \qquad \forall n \in \mathcal{N}.$$

Equivalently, the game gradient operator satisfies

$$\mathbf{v}(\mathbf{x}) = \nabla \Phi(\mathbf{x}).$$

Suppose that Φ is concave and has an L-Lipschitz gradient. Then, by the Baillon–Haddad theorem [3, Theorem 2.1.5] applied to the convex function $-\Phi$, the operator $-\nabla\Phi$ is $(1/L)$-co-coercive. Equivalently, the game is $(1/L)$-co-coercive in the sense of Definition 2.

An illustrative example is the following *concave aggregate game*, where each player benefits from increasing their own action but incurs a cost that depends on the total aggregate action across all players. Let each player $n \in \mathcal{N}$ choose an action $x_n \in \mathbb{R}^d$, and consider utilities

$$u_n(\mathbf{x}) = \langle \phi, x_n \rangle - \frac{\gamma}{2} \Big\| \sum_{m=1}^N x_m \Big\|^2, \tag{1}$$

where $\phi \in \mathbb{R}^d$ is common across players and $\gamma > 0$. The corresponding potential for the game is

$$\Phi(\mathbf{x}) = \sum_{n=1}^N \langle \phi, x_n \rangle - \frac{\gamma}{2} \Big\| \sum_{n=1}^N x_n \Big\|^2, \tag{2}$$

which is concave and smooth with $L = \gamma N$. The set of NE for this game is given by

$$\mathcal{X}^* = \left\{ \mathbf{x} \in \mathbb{R}^{Nd} \;\middle|\; \sum_{n=1}^N x_n = \frac{\phi}{\gamma} \right\},$$

which is an affine subspace of dimension $(N-1)d$.

III. Stochastic Gradient Dynamics and Noise Model

Our game evolves in discrete time. At time t, player n plays $x_{n,t}$ and obtains noisy feedback by receiving a noisy version $\hat{v}_{n,t}$ of the individual gradient evaluated at the current action profile. We denote by $\mathbf{x}_t = (x_{1,t}, \dots, x_{N,t})$ and $\hat{\mathbf{v}}_t = (\hat{v}_{1,t}, \dots, \hat{v}_{N,t})$, the joint action profile at time t and the concatenated vector of noisy gradient observations at time t, respectively.

Our players run stochastic gradient descent (SGD) using these noisy gradients to update their actions. Since players maximize utilities, this corresponds to a gradient ascent update. Formally, starting with some arbitrary profile $\mathbf{x}_0$, each player n performs the following update rule

$$x_{n,t+1} = x_{n,t} + \beta_t \hat{v}_{n,t}, \tag{3}$$

where β_t is the stepsize sequence which we discuss in detail in the next section. The noise in the gradient observation $\hat{v}_{n,t}$ can arise due to many reasons: for instance, estimates may be susceptible to random measurement errors, and the game's utility functions may be random themselves (see [4], [5]). We model the stochastic gradient noise as a martingale difference sequence, capturing unbiased but possibly state-dependent variance.

Assumption 2. *The noisy gradient $\hat{v}_{n,t}$ is of the form $\hat{v}_{n,t} = v_n(\mathbf{x}_t) + M_{n,t+1}$. Let $\mathbf{M}_t = (M_{1,t}, \dots, M_{N,t})$. Then $\mathbf{M}_{t+1}$ is a martingale difference sequence with respect to the filtration $\mathcal{F}_t = \sigma(\mathbf{x}_0, \mathbf{M}_k, k \le t)$, i.e., $\mathbb{E}[\mathbf{M}_{t+1}|\mathcal{F}_t] = 0$ for all $t \ge 0$. Moreover,*

$$\mathbb{E}[\|\mathbf{M}_{t+1}\|^2 \mid \mathcal{F}_t] \le \sigma^2(1 + \|\mathbf{x}_t\|^2).$$

Our assumption on the scaling of the martingale noise is substantially more general than the assumptions used in prior work on co-coercive games. In particular, most of the results in [2] are established under a *relative noise model*, where $\mathbb{E}[\|\mathbf{M}_{t+1}\|^2 \mid \mathcal{F}_t] \le \tau_t \|\mathbf{v}(\mathbf{x}_t)\|^2$. This assumption significantly simplifies the analysis, since the noise magnitude vanishes as the iterates approach equilibrium, i.e., as $\|\mathbf{v}(\mathbf{x}_t)\| \to 0$. Additionally, [2] assumes that (τ_t) is a decreasing sequence, which further facilitates convergence-rate analysis. They also study an *absolute noise model* of the form $\mathbb{E}[\|\mathbf{M}_{t+1}\|^2 \mid \mathcal{F}_t] \le \sigma_t^2$. Under this model, only time-average guarantees are obtained in the constant-variance case $\sigma_t = \sigma$. In contrast, we consider the more general iterate-dependent growth condition which allows the noise variance to depend on the current iterate and need not vanish near equilibrium. This framework covers both bounded-noise settings and the more general state-dependent noise models commonly assumed in stochastic optimization. Allowing the second moment of the noise to grow with the iterates is important for modeling stochastic gradient estimators on unbounded action spaces, where uniformly bounded variance assumptions are often unrealistic.

IV. Main Results

In this section, we study the convergence of the iterates $\mathbf{x}_t$ generated by (3) to the set of Nash equilibria $\mathcal{X}^*$. Our focus is on last-iterate convergence rates. Since $\mathcal{X}^* = \{\mathbf{x} : \mathbf{v}(\mathbf{x}) = 0\}$, we measure convergence through the residual $\|\mathbf{v}(\mathbf{x}_t)\|$, which captures proximity to equilibrium. Our main objective is to obtain a last-iterate mean-square bound, i.e., a bound on $\mathbb{E}[\|\mathbf{v}(\mathbf{x}_t)\|^2]$. As a consequence of our analysis, we also establish almost sure convergence of $\mathbf{x}_t$ to $\mathcal{X}^*$ and derive time-average guarantees.

We first specify the class of stepsize sequences we consider.

Assumption 3. *The step-size sequence is of the form*

$$\beta_t = \frac{1}{(t + T_0)^{\mathfrak{b}}},$$

for some $T_0 \ge \max\{1, 1/\lambda\}$ and $0.5 < \mathfrak{b} < 1$.

This choice ensures that the stepsizes are non-summable ($\sum_t \beta_t = \infty$), square-summable ($\sum_t \beta_t^2 < \infty$), and non-increasing. We do not consider step sizes of order $\Theta(1/t)$, as they are suboptimal in our setting, as discussed below. The assumption that T_0 is large enough is taken just to ensure that β_t is smaller than 1 and λ for all $t \ge 0$. This assumption is not necessary, and in absence of this assumption our result will hold for all t greater than some t_0.

Our first result establishes almost sure convergence of the iterates generated by (3) to the set of Nash equilibria.

Theorem 1. *Suppose the players update their actions according to* (3) *in a λ-co-coercive game, and that Assumptions 1–3 hold. Then the action profile $\mathbf{x}_t$ converges to the set of Nash equilibria $\mathcal{X}^*$ almost surely.*

The proof is provided in Appendix I-A. Our next result provides a bound on the time-averaged residual.

Theorem 2. *Suppose the players update their actions according to* (3) *in a λ-co-coercive game, and that Assumptions 1–3 hold. Then there exists $C_1 > 0$ such that, for all $t \ge 0$,*

$$\frac{1}{t+1} \sum_{i=0}^{t} \mathbb{E}\left[\|\mathbf{v}(\mathbf{x}_i)\|^2\right] \le \frac{C_1}{(t+1)^{1-\mathfrak{b}}}.$$

Explicit value for C_1 has been provided along with the theorem's proof in Appendix I-B. For the class of stepsize sequences in Assumption 3, the fastest time-averaged rate is achieved when $\mathfrak{b} = 0.5 + \delta$, where $\delta > 0$ can be chosen arbitrarily small. This yields a rate of $\mathcal{O}(1/t^{(0.5-\delta)})$.

We now present our main result, which provides a last-iterate mean-square bound on the residual.

Theorem 3. *Suppose the players update their actions according to* (3) *in a λ-co-coercive game, and that Assumptions 1–3 hold. Then there exist $C_2, C_3, C_4 > 0$ such that, for all $t \ge 0$,*

$$\mathbb{E}\left[\|\mathbf{v}(\mathbf{x}_t)\|^2\right] \le \begin{cases} \frac{C_2}{(t+1)^{2\mathfrak{b}-1}}, & \mathfrak{b} \in \left(\frac{1}{2}, \frac{2}{3}\right), \\ \frac{C_3 \log(t+1)}{(t+1)^{1/3}}, & \mathfrak{b} = \frac{2}{3}, \\ \frac{C_4}{(t+1)^{1-\mathfrak{b}}}, & \mathfrak{b} \in \left(\frac{2}{3}, 1\right). \end{cases}$$

The proof for this theorem has been provided in Appendix I-C. The best bound according to this result is obtained for $\mathfrak{b} = 2/3$, yielding a rate of $\mathcal{O}(\log(t)/t^{1/3})$.

V. Proof Outline

In this section, we present a proof sketch for the results in Section IV through a series of lemmas. The proofs for these lemmas have been presented in Appendix II. Our proof draws inspiration from the techniques used in the analysis of inexact Krasnosel'skii–Mann (KM) iterations [6], [7], but requires additional arguments to handle the stochastic noise.

Recall that the update rule for player n is given by $x_{n,t+1} = x_{n,t} + \beta_t \hat{v}_{n,t}$ where $\hat{v}_{n,t} = v_n(\mathbf{x}_t) + M_{n,t+1}$. Combining these updates yields the following vector form

$$\mathbf{x}_{t+1} = \mathbf{x}_t + \beta_t(\mathbf{v}(\mathbf{x}_t) + \mathbf{M}_{t+1}). \tag{4}$$

Since the game $\mathcal{G}$ is co-coercive, the operator $-\mathbf{v}(\cdot)$ is co-coercive. Our first lemma obtains an intermediate bound on $\mathbb{E}[\|\mathbf{x}_t - \mathbf{x}^*\|^2]$ where $\mathbf{x}^* \in \mathcal{X}^*$ is some NE for the game $\mathcal{G}$.

Lemma 1. *Suppose Assumptions 1-3 are satisfied. Then for all $t \geq 0$, and for some $\mathbf{x}^* \in \mathcal{X}^*$,*

$$\begin{aligned}&\mathbb{E}\left[\|\mathbf{x}_{t+1} - \mathbf{x}^*\|^2 \mid \mathcal{F}_t\right] \\ &\leq \left(1 + 2\beta_t^2\sigma^2\right)\|\mathbf{x}_t - \mathbf{x}^*\|^2 - \lambda\beta_t\|\mathbf{v}(\mathbf{x}_t)\|^2 \\ &\quad + \beta_t^2\sigma^2(1 + 2\|\mathbf{x}^*\|^2).\end{aligned}$$

The above lemma yields almost sure stability of the iterates $\{\mathbf{x}_t\}$, i.e., almost sure boundedness of the iterates, by application of the Robbins-Siegmund Theorem [8] on the recursion. This is a key requirement for the ODE approach to convergence analysis of stochastic approximation (SA) algorithms [9]. The remainder of the proof follows by rewriting iteration (4) in fixed-point form with non-expansive mappings, and invoking standard results on convergence of non-expansive SA to the set of fixed points.

Our next lemma shows that the iterates are bounded in expectation, which is essential to obtain a bound on the right hand side of the recursion.

Lemma 2. *Suppose Assumptions 1-3 hold. Then the iterates $\mathbf{x}_t$ are bounded in expectation. Specifically, there exists $\Gamma_1 > 0$ such that for all $t \geq 0$, and for $\mathbf{x}^* \in \mathcal{X}^*$,*

$$\mathbb{E}\left[\|\mathbf{x}_t - \mathbf{x}^*\|^2\right] \leq \Gamma_1.$$

Our next lemma shows that the sum of weighted residual errors is bounded by a constant.

Lemma 3. *Suppose Assumptions 1-3 hold. Then there exists Γ_2 such that for all $t \geq 0$,*

$$\sum_{i=0}^{t} \beta_i \mathbb{E}\left[\|\mathbf{v}(\mathbf{x}_i)\|^2\right] \leq \Gamma_2.$$

The time-average guarantee (Theorem 2) follows directly from this lemma by simply noting that β_i is non-increasing, which implies that

$$\beta_t \sum_{i=0}^{t} \mathbb{E}\left[\|\mathbf{v}(\mathbf{x}_i)\|^2\right] \leq \sum_{i=0}^{t} \beta_i \mathbb{E}\left[\|\mathbf{v}(\mathbf{x}_i)\|^2\right] \leq \Gamma_2.$$

But our primary goal is to obtain a last-iterate bound, i.e., obtain a bound on $\mathbb{E}[\|\mathbf{v}(\mathbf{x}_t)\|^2]$. For this, we would like to get a bound on $\mathbb{E}[\|\mathbf{v}(\mathbf{x}_i)\|^2]$ in terms of $\mathbb{E}[\|\mathbf{v}(\mathbf{x}_t)\|^2]$. But the martingale difference noise $\mathbf{M}_{t+1}$ does not allow us to obtain a useful bound directly. To solve this, we define an averaged error sequence $\mathbf{U}_{t+1} = (1 - \beta_t)\mathbf{U}_t + \beta_t\mathbf{M}_{t+1}$, with $\mathbf{U}_0 = 0$, and a modified iterate $\mathbf{z}_t = \mathbf{x}_t - \mathbf{U}_t$. We first show that $\mathbb{E}[\|\mathbf{x}_t - \mathbf{z}_t\|^2]$ decays at a sufficiently fast rate, implying that we can analyze $\mathbb{E}[\|\mathbf{v}(\mathbf{z}_t)\|^2]$ instead of $\mathbb{E}[\|\mathbf{v}(\mathbf{x}_t)\|^2]$. The next step is studying the iterates $\mathbf{z}_t$ and obtaining an iteration for these iterates. And finally, we fulfill the purpose of studying $\mathbf{z}_t$ in terms of $\mathbf{x}_t$, by obtaining a bound on $\mathbb{E}[\|\mathbf{v}(z_i)\|^2]$ in terms of $\mathbb{E}[\|\mathbf{v}(z_t)\|^2]$.

Lemma 4. *Suppose Assumptions 1-3 hold.*

a) There exists $\Gamma_3 > 0$ such that for all $t \geq 0$,

$$\mathbb{E}\left[\|\mathbf{x}_t - \mathbf{z}_t\|^2\right] = \mathbb{E}\left[\|\mathbf{U}_t\|^2\right] \leq \Gamma_3\beta_t.$$

b) There exists $\Gamma_4 > 0$ such that for all $t \geq 0$,

$$\sum_{i=0}^{t} \beta_i \mathbb{E}\left[\|\mathbf{v}(\mathbf{z}_i)\|^2\right] \leq \Gamma_4.$$

c) For all $i \leq t - 1$, there exists $\Gamma_5 > 0$ such that

$$\mathbb{E}\left[\|\mathbf{v}(\mathbf{z}_i)\|^2\right] \geq \mathbb{E}\left[\|\mathbf{v}(\mathbf{z}_t)\|^2\right] - \Gamma_5 \sum_{j=i}^{t-1} \beta_j^2.$$

The above lemma implies that

$$\mathbb{E}\left[\|\mathbf{v}(\mathbf{z}_t)\|^2\right] \sum_{i=0}^{t} \beta_i \leq \Gamma_4 + \Gamma_5 \sum_{i=0}^{t} \beta_i \sum_{j=i}^{t-1} \beta_j^2.$$

The summation on the right hand side is bounded for $\mathfrak{b} > 2/3$, $\mathcal{O}(\log(t))$ for $\mathfrak{b} = 2/3$, and $\mathcal{O}(t^{2-3b})$ for $1/2 < b < 2/3$. This completes the proof for the last-iterate bound.

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APPENDIX I
PROOFS FOR THEOREMS 1, 2, 3

A. Proof for Theorem 1

Proof. Define $V_t = \|\mathbf{x}_t - \mathbf{x}^*\|^2$, $A_t = 2\beta_t^2\sigma^2$, $B_t = \lambda\beta_t\|\mathbf{v}(\mathbf{x}_t)\|^2$, and $C_t = \beta_t^2\sigma^2(1+2\|\mathbf{x}^*\|^2)$. Using Lemma 1, we get

$$\mathbb{E}[V_{t+1}|\mathcal{F}_t] \le (1+A_t)V_t - B_t + C_t.$$

Since β_t is square-summable (Assumption 3), we have $\sum_t A_t < \infty$ and $\sum_t C_t < \infty$. Then, we can apply Robbins-Siegmund Theorem [8] which says that V_t converges to a finite random variable. In particular, $\sup_t V_t < \infty$ almost surely, and hence

$$\sup_t \|\mathbf{x}_t\| < \infty \qquad \text{a.s.}$$

Thus, the iterates are almost surely bounded.

Now the iteration in (4) can be written as

$$\mathbf{x}_{t+1} = \mathbf{x}_t + \alpha_t(h(\mathbf{x}_t) - \mathbf{x}_t + \overline{\mathbf{M}}_{t+1}), \tag{5}$$

where $\alpha_t = \frac{\beta_t}{2\lambda} = \frac{1/(2\lambda)}{(t+T_0)^{\mathfrak{b}}}$, $h(\mathbf{x}) = \mathbf{x} + 2\lambda\mathbf{v}(\mathbf{x})$, and $\overline{\mathbf{M}}_{t+1} = 2\lambda\mathbf{M}_{t+1}$. Here $h(\cdot)$ is non-expansive [10, Proposition 4.11], i.e., $h(\cdot)$ is Lipschitz with constant 1. This implies that the players' update rule can be viewed as a fixed-point iteration with a non-expansive mapping. Moreover, by definition, the fixed points for $h(\cdot)$ are the NE of the game, i.e., $\mathcal{X}^* = \{\mathbf{x}|h(\mathbf{x}) = \mathbf{x}\}$. Since the iterates are almost surely bounded, we can study the ODE limit associated with (5):

$$\dot{\mathbf{x}}(t) = h(\mathbf{x}(t)) - \mathbf{x}(t).$$

Then, Corollary 12.1 and Remark 3 below Theorem 12.2 from [9] together imply that $\mathbf{x}_t$ converges almost surely to the set $\mathcal{X}^*$, completing the proof for Theorem 1. □

B. Proof for Theorem 2

Proof. By Lemma 3,

$$\sum_{i=0}^{t} \beta_i \mathbb{E}\left[\|\mathbf{v}(\mathbf{x}_i)\|^2\right] \le \Gamma_2, \qquad \forall t \ge 0.$$

Since $\{\beta_t\}$ is non-increasing,

$$\beta_t \sum_{i=0}^{t} \mathbb{E}\left[\|\mathbf{v}(\mathbf{x}_i)\|^2\right] \le \sum_{i=0}^{t} \beta_i \mathbb{E}\left[\|\mathbf{v}(\mathbf{x}_i)\|^2\right] \le \Gamma_2.$$

Therefore,

$$\frac{1}{t+1}\sum_{i=0}^{t} \mathbb{E}\left[\|\mathbf{v}(\mathbf{x}_i)\|^2\right] \le \frac{\Gamma_2}{(t+1)\beta_t}.$$

Using Assumption 3,

$$\beta_t = \frac{1}{(t+T_0)^{\mathfrak{b}}},$$

and hence

$$\frac{\Gamma_2}{(t+1)\beta_t} = \Gamma_2 \frac{(t+T_0)^{\mathfrak{b}}}{t+1}.$$

Since

$$t + T_0 \le \max\{1, T_0\}(t+1),$$

we obtain

$$\frac{\Gamma_2}{(t+1)\beta_t} \le \Gamma_2 \max\{1, T_0\}^{\mathfrak{b}}(t+1)^{-(1-\mathfrak{b})}.$$

Thus the conclusion follows with

$$C_1 := \Gamma_2 \max\{1, T_0\}^{\mathfrak{b}}.$$

□

C. Proof for Theorem 3

Proof. For each $i \le t-1$, Lemma 4(c) implies

$$\mathbb{E}\left[\|\mathbf{v}(\mathbf{z}_t)\|^2\right] \le \mathbb{E}\left[\|\mathbf{v}(\mathbf{z}_i)\|^2\right] + \Gamma_5 \sum_{j=i}^{t-1} \beta_j^2.$$

Multiplying by β_i and summing from $i = 0$ to $t-1$, we get

$$\mathbb{E}\left[\|\mathbf{v}(\mathbf{z}_t)\|^2\right] \sum_{i=0}^{t-1} \beta_i \le \sum_{i=0}^{t-1} \beta_i \mathbb{E}\left[\|\mathbf{v}(\mathbf{z}_i)\|^2\right] + \Gamma_5 \sum_{i=0}^{t-1} \beta_i \sum_{j=i}^{t-1} \beta_j^2.$$

Adding the nonnegative term $\beta_t\mathbb{E}[\|\mathbf{v}(\mathbf{z}_t)\|^2]$ to both sides and using Lemma 4(b) yields

$$\mathbb{E}\left[\|\mathbf{v}(\mathbf{z}_t)\|^2\right] \sum_{i=0}^{t} \beta_i \le \Gamma_4 + \Gamma_5 \sum_{i=0}^{t-1} \beta_i \sum_{j=i}^{t-1} \beta_j^2. \tag{6}$$

From (6), we directly obtain

$$\mathbb{E}\left[\|\mathbf{v}(\mathbf{z}_t)\|^2\right] \le \frac{\Gamma_4}{\sum_{i=0}^{t}\beta_i} + \Gamma_5 \frac{\sum_{i=0}^{t-1}\beta_i \sum_{j=i}^{t-1}\beta_j^2}{\sum_{i=0}^{t}\beta_i}.$$

Since $\beta_i = (i+T_0)^{-\mathfrak{b}}$ with $\mathfrak{b} \in (1/2, 1)$ and T_0 is fixed, standard integral comparisons give

$$\sum_{i=0}^{t} \beta_i = \Theta\big((t+1)^{1-\mathfrak{b}}\big),$$

and

$$\sum_{i=0}^{t-1} \beta_i \sum_{j=i}^{t-1} \beta_j^2 = \sum_{j=0}^{t-1}\Big(\sum_{i=0}^{j}\beta_i\Big)\beta_j^2 = \begin{cases} \mathcal{O}\big((t+1)^{2-3\mathfrak{b}}\big), & \mathfrak{b} \in \big(\frac{1}{2}, \frac{2}{3}\big), \\ \mathcal{O}\big(\log(t+2)\big), & \mathfrak{b} = \frac{2}{3}, \\ \mathcal{O}(1), & \mathfrak{b} \in \big(\frac{2}{3}, 1\big). \end{cases}$$

Therefore,

$$\frac{\Gamma_4}{\sum_{i=0}^{t}\beta_i} = \mathcal{O}\big((t+1)^{-(1-\mathfrak{b})}\big),$$

and

$$\Gamma_5 \frac{\sum_{i=0}^{t-1} \beta_i \sum_{j=i}^{t-1} \beta_j^2}{\sum_{i=0}^{t} \beta_i} \leq \begin{cases} \mathcal{O}\big((t+1)^{-(2\mathfrak{b}-1)}\big), & \mathfrak{b} \in \big(\frac{1}{2}, \frac{2}{3}\big), \\ \mathcal{O}\big((t+1)^{-1/3} \log(t+2)\big), & \mathfrak{b} = \frac{2}{3}, \\ \mathcal{O}\big((t+1)^{-(1-\mathfrak{b})}\big), & \mathfrak{b} \in \big(\frac{2}{3}, 1\big). \end{cases}$$

Combining the two bounds, we get

$$\mathbb{E}\left[\|\mathbf{v}(\mathbf{z}_t)\|^2\right] \leq \begin{cases} \mathcal{O}\big((t+1)^{-(2\mathfrak{b}-1)}\big), & \mathfrak{b} \in \big(\frac{1}{2}, \frac{2}{3}\big), \\ \mathcal{O}\big((t+1)^{-1/3} \log(t+2)\big), & \mathfrak{b} = \frac{2}{3}, \\ \mathcal{O}\big((t+1)^{-(1-\mathfrak{b})}\big), & \mathfrak{b} \in \big(\frac{2}{3}, 1\big). \end{cases}$$

Next, we transfer the bound from $\mathbf{z}_t$ to $\mathbf{x}_t$. By the $(1/\lambda)$-Lipschitz property of $\mathbf{v}$,

$$\|\mathbf{v}(\mathbf{x}_t)\| \leq \|\mathbf{v}(\mathbf{z}_t)\| + \|\mathbf{v}(\mathbf{x}_t) - \mathbf{v}(\mathbf{z}_t)\| \leq \|\mathbf{v}(\mathbf{z}_t)\| + \frac{1}{\lambda}\|\mathbf{U}_t\|.$$

Hence

$$\|\mathbf{v}(\mathbf{x}_t)\|^2 \leq 2\|\mathbf{v}(\mathbf{z}_t)\|^2 + \frac{2}{\lambda^2}\|\mathbf{U}_t\|^2.$$

Taking expectation and using Lemma 4(a), we get

$$\mathbb{E}\left[\|\mathbf{v}(\mathbf{x}_t)\|^2\right] \leq 2\mathbb{E}\left[\|\mathbf{v}(\mathbf{z}_t)\|^2\right] + \frac{2\Gamma_3}{\lambda^2}\beta_t.$$

Since

$$\beta_t = (t+T_0)^{-\mathfrak{b}} = \mathcal{O}\big((t+1)^{-\mathfrak{b}}\big),$$

the second term is of smaller order than the main rate in each of the three regimes:

$$(t+1)^{-\mathfrak{b}} = \mathcal{O}\big((t+1)^{-(2\mathfrak{b}-1)}\big) \quad \text{for } \mathfrak{b} \in \left(\frac{1}{2}, \frac{2}{3}\right),$$

$$(t+1)^{-2/3} = \mathcal{O}\big((t+1)^{-1/3} \log(t+1)\big),$$

and

$$(t+1)^{-\mathfrak{b}} = \mathcal{O}\big((t+1)^{-(1-\mathfrak{b})}\big) \quad \text{for } \mathfrak{b} \in \left(\frac{2}{3}, 1\right).$$

Thus, there exist $C_2, C_3, C_4 > 0$ such that

$$\mathbb{E}\left[\|\mathbf{v}(\mathbf{x}_t)\|^2\right] \leq \begin{cases} \frac{C_2}{(t+1)^{2\mathfrak{b}-1}}, & \mathfrak{b} \in \big(\frac{1}{2}, \frac{2}{3}\big), \\ \frac{C_3 \log(t+2)}{(t+1)^{1/3}}, & \mathfrak{b} = \frac{2}{3}, \\ \frac{C_4}{(t+1)^{1-\mathfrak{b}}}, & \mathfrak{b} \in \big(\frac{2}{3}, 1\big). \end{cases}$$

This proves the theorem. □

APPENDIX II
PROOFS FROM SECTION V

A. Proof for Lemma 1

Proof. Let $\mathbf{x}^*$ be such that $\mathbf{v}(\mathbf{x}^*) = 0$. Then, for all $t \geq 0$,

$$\mathbf{x}_{t+1} - \mathbf{x}^* = \mathbf{x}_t - \mathbf{x}^* + \beta_t(\mathbf{v}(\mathbf{x}_t) - \mathbf{v}(\mathbf{x}^*) + \mathbf{M}_{t+1}).$$

Now, using the standard expansion, we get

$$\begin{aligned}\|\mathbf{x}_{t+1} - \mathbf{x}^*\|^2 &= \|\mathbf{x}_t - \mathbf{x}^* + \beta_t(\mathbf{v}(\mathbf{x}_t) - \mathbf{v}(\mathbf{x}^*) + \mathbf{M}_{t+1})\|^2 \\ &= \|\mathbf{x}_t - \mathbf{x}^*\|^2 + \beta_t^2\|\mathbf{v}(\mathbf{x}_t) - \mathbf{v}(\mathbf{x}^*) + \mathbf{M}_{t+1}\|^2 \qquad \text{(7a)} \\ &\quad + 2\beta_t\langle \mathbf{x}_t - \mathbf{x}^*, \mathbf{v}(\mathbf{x}_t) - \mathbf{v}(\mathbf{x}^*) + \mathbf{M}_{t+1}\rangle. \qquad \text{(7b)}\end{aligned}$$

For the term (7a), we note that

$$\|\mathbf{v}(\mathbf{x}_t) - \mathbf{v}(\mathbf{x}^*) + \mathbf{M}_{t+1}\|^2 = \|\mathbf{v}(\mathbf{x}_t) - \mathbf{v}(\mathbf{x}^*)\|^2 + \|\mathbf{M}_{t+1}\|^2 + 2\langle \mathbf{v}(\mathbf{x}_t) - \mathbf{v}(\mathbf{x}^*), \mathbf{M}_{t+1}\rangle.$$

For the term (7b), since $-\mathbf{v}(\cdot)$ is λ-co-coercive, we have

$$\langle \mathbf{x}_t - \mathbf{x}^*, \mathbf{v}(\mathbf{x}_t) - \mathbf{v}(\mathbf{x}^*)\rangle \leq -\lambda\|\mathbf{v}(\mathbf{x}_t) - \mathbf{v}(\mathbf{x}^*)\|^2.$$

Hence,

$$2\beta_t\langle \mathbf{x}_t - \mathbf{x}^*, \mathbf{v}(\mathbf{x}_t) - \mathbf{v}(\mathbf{x}^*) + \mathbf{M}_{t+1}\rangle \leq -2\lambda\beta_t\|\mathbf{v}(\mathbf{x}_t) - \mathbf{v}(\mathbf{x}^*)\|^2 + 2\beta_t\langle \mathbf{x}_t - \mathbf{x}^*, \mathbf{M}_{t+1}\rangle.$$

Returning to (7), we get

$$\begin{aligned}\|\mathbf{x}_{t+1} - \mathbf{x}^*\|^2 &\leq \|\mathbf{x}_t - \mathbf{x}^*\|^2 + \beta_t^2\|\mathbf{v}(\mathbf{x}_t) - \mathbf{v}(\mathbf{x}^*)\|^2 + \beta_t^2\|\mathbf{M}_{t+1}\|^2 \\ &\quad + 2\beta_t^2\langle \mathbf{v}(\mathbf{x}_t) - \mathbf{v}(\mathbf{x}^*), \mathbf{M}_{t+1}\rangle - 2\lambda\beta_t\|\mathbf{v}(\mathbf{x}_t) - \mathbf{v}(\mathbf{x}^*)\|^2 \\ &\quad + 2\beta_t\langle \mathbf{x}_t - \mathbf{x}^*, \mathbf{M}_{t+1}\rangle \\ &= \|\mathbf{x}_t - \mathbf{x}^*\|^2 - \beta_t(2\lambda - \beta_t)\|\mathbf{v}(\mathbf{x}_t) - \mathbf{v}(\mathbf{x}^*)\|^2 + \beta_t^2\|\mathbf{M}_{t+1}\|^2 \\ &\quad + 2\beta_t\langle \mathbf{x}_t - \mathbf{x}^*, \mathbf{M}_{t+1}\rangle + 2\beta_t^2\langle \mathbf{v}(\mathbf{x}_t) - \mathbf{v}(\mathbf{x}^*), \mathbf{M}_{t+1}\rangle.\end{aligned}$$

Since $T_0 \geq 1/\lambda$, $\beta_t \leq \lambda$ for all t, then $2\lambda - \beta_t \geq \lambda$, and so

$$\begin{aligned}\|\mathbf{x}_{t+1} - \mathbf{x}^*\|^2 &\leq \|\mathbf{x}_t - \mathbf{x}^*\|^2 - \lambda\beta_t\|\mathbf{v}(\mathbf{x}_t) - \mathbf{v}(\mathbf{x}^*)\|^2 + \beta_t^2\|\mathbf{M}_{t+1}\|^2 \\ &\quad + 2\beta_t\langle \mathbf{x}_t - \mathbf{x}^*, \mathbf{M}_{t+1}\rangle + 2\beta_t^2\langle \mathbf{v}(\mathbf{x}_t) - \mathbf{v}(\mathbf{x}^*), \mathbf{M}_{t+1}\rangle.\end{aligned}$$

On taking expectation conditioned on $\mathcal{F}_t$, we have

$$\mathbb{E}[\langle \mathbf{x}_t - \mathbf{x}^*, \mathbf{M}_{t+1}\rangle \mid \mathcal{F}_t] = 0, \qquad \mathbb{E}[\langle \mathbf{v}(\mathbf{x}_t) - \mathbf{v}(\mathbf{x}^*), \mathbf{M}_{t+1}\rangle \mid \mathcal{F}_t] = 0,$$

since $\{\mathbf{M}_{t+1}\}$ is a martingale difference sequence with respect to $\{\mathcal{F}_t\}$, and both $\mathbf{x}_t$ and $\mathbf{v}(\mathbf{x}_t)$ are $\mathcal{F}_t$-measurable.

Therefore,

$$\begin{aligned}\mathbb{E}\left[\|\mathbf{x}_{t+1} - \mathbf{x}^*\|^2 \mid \mathcal{F}_t\right] &\leq \|\mathbf{x}_t - \mathbf{x}^*\|^2 + \beta_t^2\mathbb{E}\left[\|\mathbf{M}_{t+1}\|^2 \mid \mathcal{F}_t\right] - \lambda\beta_t\|\mathbf{v}(\mathbf{x}_t) - \mathbf{v}(\mathbf{x}^*)\|^2 \\ &\leq \|\mathbf{x}_t - \mathbf{x}^*\|^2 + \beta_t^2\sigma^2(1 + \|\mathbf{x}_t\|^2) - \lambda\beta_t\|\mathbf{v}(\mathbf{x}_t)\|^2 \\ &\leq \|\mathbf{x}_t - \mathbf{x}^*\|^2 + 2\beta_t^2\sigma^2\|\mathbf{x}_t - \mathbf{x}^*\|^2 + \beta_t^2\sigma^2(1 + 2\|\mathbf{x}^*\|^2) - \lambda\beta_t\|\mathbf{v}(\mathbf{x}_t)\|^2,\end{aligned}$$

where we used $\mathbf{v}(\mathbf{x}^*) = 0$ for the second inequality. This completes the proof for Lemma 1.

□

B. Proof for Lemma 2

Proof. Taking expectation and dropping the negative residual term in Lemma 1 gives, for all $t \geq 0$,

$$\mathbb{E}\left[\|\mathbf{x}_{t+1} - \mathbf{x}^*\|^2\right] \leq \mathbb{E}\left[\|\mathbf{x}_t - \mathbf{x}^*\|^2\right] + \beta_t^2\sigma^2(1 + 2\|\mathbf{x}^*\|^2) + 2\beta_t^2\sigma^2\mathbb{E}\left[\|\mathbf{x}_t - \mathbf{x}^*\|^2\right].$$

Summing from $i = 0$ to $t - 1$, we get

$$\mathbb{E}\left[\|\mathbf{x}_t - \mathbf{x}^*\|^2\right] \leq \mathbb{E}\left[\|\mathbf{x}_0 - \mathbf{x}^*\|^2\right] + \sigma^2\left(1 + 2\|\mathbf{x}^*\|^2\right)\sum_{i=0}^{t-1}\beta_i^2 + 2\sigma^2\sum_{i=0}^{t-1}\beta_i^2\mathbb{E}\left[\|\mathbf{x}_i - \mathbf{x}^*\|^2\right]$$

Set

$$B = \mathbb{E}\left[\|\mathbf{x}_0 - \mathbf{x}^*\|^2\right] + \sigma^2\big(1 + 2\|\mathbf{x}^*\|^2\big)\sum_{i=0}^{\infty}\beta_i^2.$$

Then

$$\mathbb{E}\left[\|\mathbf{x}_t - \mathbf{x}^*\|^2\right] \le B + 2\sigma^2\sum_{i=0}^{t-1}\beta_i^2\mathbb{E}\left[\|\mathbf{x}_i - \mathbf{x}^*\|^2\right].$$

By the discrete Grönwall inequality it follows that

$$\mathbb{E}\left[\|\mathbf{x}_t - \mathbf{x}^*\|^2\right] \le B\exp\Big(2\sigma^2\sum_{i=0}^{t-1}\beta_i^2\Big),$$

Since $\sum_{i=0}^{\infty}\beta_i^2 < \infty$, the right-hand side of the above bound is itself bounded by some constant $\Gamma_1 := B\exp\Big(2\sigma^2\sum_{i=0}^{\infty}\beta_i^2\Big) > 0$. Hence there exists $\Gamma_1 > 0$ such that

$$\sup_{t\ge 0}\mathbb{E}\left[\|\mathbf{x}_t - \mathbf{x}^*\|^2\right] \le \Gamma_1.$$

□

C. Proof for Lemma 3

Proof. Taking expectation in Lemma 1, and rearranging terms, we get

$$\lambda\beta_t\mathbb{E}\left[\|\mathbf{v}(\mathbf{x}_t)\|^2\right] \le \mathbb{E}\left[\|\mathbf{x}_t - \mathbf{x}^*\|^2\right] - \mathbb{E}\left[\|\mathbf{x}_{t+1} - \mathbf{x}^*\|^2\right] + \beta_t^2\sigma^2\big(1 + 2\|\mathbf{x}^*\|^2\big) + 2\beta_t^2\sigma^2\mathbb{E}\left[\|\mathbf{x}_t - \mathbf{x}^*\|^2\right].$$

Summing from $i = 0$ to t gives

$$\lambda\sum_{i=0}^{t}\beta_i\mathbb{E}\left[\|\mathbf{v}(\mathbf{x}_i)\|^2\right] \le \mathbb{E}\left[\|\mathbf{x}_0 - \mathbf{x}^*\|^2\right] - \mathbb{E}\left[\|\mathbf{x}_{t+1} - \mathbf{x}^*\|^2\right] + \sigma^2\big(1 + 2\|\mathbf{x}^*\|^2\big)\sum_{i=0}^{t}\beta_i^2 + 2\sigma^2\sum_{i=0}^{t}\beta_i^2\mathbb{E}\left[\|\mathbf{x}_i - \mathbf{x}^*\|^2\right].$$

Dropping the nonnegative term $\mathbb{E}[\|\mathbf{x}_{t+1} - \mathbf{x}^*\|^2]$, we obtain

$$\lambda\sum_{i=0}^{t}\beta_i\mathbb{E}\left[\|\mathbf{v}(\mathbf{x}_i)\|^2\right] \le \mathbb{E}\left[\|\mathbf{x}_0 - \mathbf{x}^*\|^2\right] + \sigma^2\big(1 + 2\|\mathbf{x}^*\|^2\big)\sum_{i=0}^{t}\beta_i^2 + 2\sigma^2\sum_{i=0}^{t}\beta_i^2\mathbb{E}\left[\|\mathbf{x}_i - \mathbf{x}^*\|^2\right].$$

Now using

$$\mathbb{E}\left[\|\mathbf{x}_i - \mathbf{x}^*\|^2\right] \le \Gamma_1,$$

we get

$$\lambda\sum_{i=0}^{t}\beta_i\mathbb{E}\left[\|\mathbf{v}(\mathbf{x}_i)\|^2\right] \le \mathbb{E}\left[\|\mathbf{x}_0 - \mathbf{x}^*\|^2\right] + \sigma^2\big(1 + 2\|\mathbf{x}^*\|^2\big)\sum_{i=0}^{t}\beta_i^2 + 2\sigma^2\Gamma_1\sum_{i=0}^{t}\beta_i^2.$$

Since $\sum_{i=0}^{\infty}\beta_i^2 < \infty$, we have

$$\lambda\sum_{i=0}^{t}\beta_i\mathbb{E}\left[\|\mathbf{v}(\mathbf{x}_i)\|^2\right] \le \mathbb{E}\left[\|\mathbf{x}_0 - \mathbf{x}^*\|^2\right] + \Big(\sigma^2\big(1 + 2\|\mathbf{x}^*\|^2\big) + 2\sigma^2\Gamma_1\Big)\sum_{i=0}^{\infty}\beta_i^2.$$

Define

$$\Gamma_2 := \frac{1}{\lambda}\left(\mathbb{E}\left[\|\mathbf{x}_0 - \mathbf{x}^*\|^2\right] + \Big(\sigma^2\big(1 + 2\|\mathbf{x}^*\|^2\big) + 2\sigma^2\Gamma_1\Big)\sum_{i=0}^{\infty}\beta_i^2\right).$$

Then, for all $t \ge 0$,

$$\sum_{i=0}^{t}\beta_i\mathbb{E}\left[\|\mathbf{v}(\mathbf{x}_i)\|^2\right] \le \Gamma_2.$$

This completes the proof. □

D. Proof for Lemma 4

Proof. We first note that co-coercivity of $-\mathbf{v}$ implies that $\mathbf{v}$ is $(1/\lambda)$-Lipschitz. Indeed, for all $\mathbf{x}, \mathbf{x}' \in \mathbb{R}^{Nd}$,

$$\lambda\|\mathbf{v}(\mathbf{x}) - \mathbf{v}(\mathbf{x}')\|^2 \le -\langle \mathbf{x} - \mathbf{x}', \mathbf{v}(\mathbf{x}) - \mathbf{v}(\mathbf{x}')\rangle \le \|\mathbf{x} - \mathbf{x}'\|\,\|\mathbf{v}(\mathbf{x}) - \mathbf{v}(\mathbf{x}')\|,$$

and hence

$$\|\mathbf{v}(\mathbf{x}) - \mathbf{v}(\mathbf{x}')\| \le \frac{1}{\lambda}\|\mathbf{x} - \mathbf{x}'\|.$$

(a) Bound for $\mathbb{E}\|\mathbf{U}_k\|^2$.

Expanding and taking conditional expectation yields

$$\begin{aligned}
\|\mathbf{U}_{k+1}\|^2 &= \|(1-\beta_k)\mathbf{U}_k + \beta_k \mathbf{M}_{k+1}\|_2^2 \\
&= (1-\beta_k)^2\|\mathbf{U}_k\|^2 + 2(1-\beta_k)\beta_k\langle \mathbf{U}_k, \mathbf{M}_{k+1}\rangle + \beta_k^2\|\mathbf{M}_{k+1}\|^2, \\
\mathbb{E}[\|\mathbf{U}_{k+1}\|^2 \mid \mathcal{F}_k] &= (1-\beta_k)^2\|\mathbf{U}_k\|^2 + \beta_k^2\,\mathbb{E}[\|\mathbf{M}_{k+1}\|^2 \mid \mathcal{F}_k],
\end{aligned}$$

since $\mathbb{E}[\mathbf{M}_{k+1} \mid \mathcal{F}_k] = 0$ implies

$$\mathbb{E}[\langle \mathbf{U}_k, \mathbf{M}_{k+1}\rangle \mid \mathcal{F}_k] = \langle \mathbf{U}_k, \mathbb{E}[\mathbf{M}_{k+1} \mid \mathcal{F}_k]\rangle = 0.$$

Taking total expectation and using

$$\mathbb{E}\|\mathbf{M}_{k+1}\|^2 \le \sigma^2\big(1 + \mathbb{E}\|\mathbf{x}_k\|^2\big) \le \sigma^2\big(1 + 2\mathbb{E}\|\mathbf{x}_k - \mathbf{x}^*\|^2 + 2\|\mathbf{x}^*\|^2\big) \le \sigma^2\big(1 + 2\Gamma_1 + 2\|\mathbf{x}^*\|^2\big) =: A,$$

we get

$$\mathbb{E}\|\mathbf{U}_{k+1}\|^2 \le (1-\beta_k)^2\mathbb{E}\|\mathbf{U}_k\|^2 + A\beta_k^2.$$

Since $\beta_k \in (0,1]$, we have $(1-\beta_k)^2 \le (1-\beta_k)$, hence

$$\mathbb{E}\|\mathbf{U}_{k+1}\|^2 \le (1-\beta_k)\mathbb{E}\|\mathbf{U}_k\|^2 + A\beta_k^2. \tag{8}$$

Now we know,

$$\beta_k = \frac{1}{(k+T_0)^{\mathfrak{b}}}, \qquad 0.5 < \mathfrak{b} < 1.$$

Hence

$$\beta_k - \beta_{k+1} = \frac{1}{(k+T_0)^{\mathfrak{b}}} - \frac{1}{(k+1+T_0)^{\mathfrak{b}}} = \frac{1}{(k+T_0)^{\mathfrak{b}}}\Big(1 - \Big(1 + \frac{1}{k+T_0}\Big)^{-\mathfrak{b}}\Big).$$

Using the inequality $1 - (1+t)^{-\mathfrak{b}} \le \mathfrak{b}t$ for $t \ge 0$, we obtain

$$\beta_k - \beta_{k+1} \le \frac{1}{(k+T_0)^{\mathfrak{b}}} \cdot \frac{\mathfrak{b}}{k+T_0} = \frac{\mathfrak{b}}{(k+T_0)^{\mathfrak{b}+1}}.$$

Also,

$$\beta_k^2 = \frac{1}{(k+T_0)^{2\mathfrak{b}}}.$$

Therefore

$$\frac{\beta_k - \beta_{k+1}}{\beta_k^2} \le \frac{\mathfrak{b}}{(k+T_0)^{1-\mathfrak{b}}} \to 0.$$

Since $\mathfrak{b} < 1$, there exists k_0 such that

$$\beta_k - \beta_{k+1} \le \frac{1}{2}\beta_k^2, \qquad \forall k \ge k_0.$$

Now define

$$\Gamma_3 := \max\Big\{2A,\ \max_{0\le k\le k_0} \frac{\mathbb{E}\|\mathbf{U}_k\|^2}{\beta_k}\Big\}.$$

Then, by definition,

$$\mathbb{E}\|\mathbf{U}_k\|^2 \le \Gamma_3\beta_k, \qquad \forall\, 0 \le k \le k_0.$$

Now fix $k \ge k_0$ and assume inductively that

$$\mathbb{E}\|\mathbf{U}_k\|^2 \le \Gamma_3\beta_k.$$

Then by (8),

$$\mathbb{E}\|\mathbf{U}_{k+1}\|^2 \le (1-\beta_k)\Gamma_3\beta_k + A\beta_k^2 = \Gamma_3\beta_k - (\Gamma_3 - A)\beta_k^2.$$

Since $\Gamma_3 \geq 2A$, we have $\Gamma_3 - A \geq \Gamma_3/2$, and therefore

$$\mathbb{E}\|\mathbf{U}_{k+1}\|^2 \leq \Gamma_3\beta_k - \frac{\Gamma_3}{2}\beta_k^2.$$

Also, since $k \geq k_0$,

$$\beta_k - \beta_{k+1} \leq \frac{1}{2}\beta_k^2,$$

which implies

$$\Gamma_3(\beta_k - \beta_{k+1}) \leq \frac{\Gamma_3}{2}\beta_k^2.$$

Hence

$$\mathbb{E}\|\mathbf{U}_{k+1}\|^2 \leq \Gamma_3\beta_k - \Gamma_3(\beta_k - \beta_{k+1}) = \Gamma_3\beta_{k+1}.$$

Thus, by induction,

$$\mathbb{E}\|\mathbf{U}_k\|^2 \leq \Gamma_3\beta_k, \qquad \forall k \geq 0.$$

This completes the proof for part (a).

(b) Bound on the summation of $\beta_i\mathbb{E}[\|\mathbf{v}(\mathbf{z}_i)\|^2]$.
Using the Lipschitz property of $\mathbf{v}$,

$$\|\mathbf{v}(\mathbf{z}_t)\| \leq \|\mathbf{v}(\mathbf{x}_t)\| + \|\mathbf{v}(\mathbf{z}_t) - \mathbf{v}(\mathbf{x}_t)\| \leq \|\mathbf{v}(\mathbf{x}_t)\| + \frac{1}{\lambda}\|\mathbf{U}_t\|.$$

Hence

$$\|\mathbf{v}(\mathbf{z}_t)\|^2 \leq 2\|\mathbf{v}(\mathbf{x}_t)\|^2 + \frac{2}{\lambda^2}\|\mathbf{U}_t\|^2.$$

Multiplying by β_t, taking expectation, and summing from $i = 0$ to t, we get

$$\begin{aligned}\sum_{i=0}^{t}\beta_i\mathbb{E}\left[\|\mathbf{v}(\mathbf{z}_i)\|^2\right] &\leq 2\sum_{i=0}^{t}\beta_i\mathbb{E}\left[\|\mathbf{v}(\mathbf{x}_i)\|^2\right] + \frac{2}{\lambda^2}\sum_{i=0}^{t}\beta_i\mathbb{E}\left[\|\mathbf{U}_i\|^2\right] \\ &\leq 2\Gamma_2 + \frac{2\Gamma_3}{\lambda^2}\sum_{i=0}^{\infty}\beta_i^2.\end{aligned}$$

Define

$$\Gamma_4 := 2\Gamma_2 + \frac{2\Gamma_3}{\lambda^2}\sum_{i=0}^{\infty}\beta_i^2.$$

Then

$$\sum_{i=0}^{t}\beta_i\mathbb{E}\left[\|\mathbf{v}(\mathbf{z}_i)\|^2\right] \leq \Gamma_4, \qquad \forall t \geq 0.$$

(c) Bound on $\mathbb{E}[\|\mathbf{v}(\mathbf{z}_i)\|^2]$ in terms of $\mathbb{E}[\|\mathbf{v}(\mathbf{z}_t)\|^2]$.
Subtracting the recursion for $\mathbf{U}_t$ from the recursion

$$\mathbf{x}_{t+1} = \mathbf{x}_t + \beta_t\big(\mathbf{v}(\mathbf{x}_t) + \mathbf{M}_{t+1}\big)$$

gives

$$\begin{aligned}\mathbf{z}_{t+1} &= \mathbf{x}_{t+1} - \mathbf{U}_{t+1} \\ &= \mathbf{x}_t + \beta_t\big(\mathbf{v}(\mathbf{x}_t) + \mathbf{M}_{t+1}\big) - \big((1-\beta_t)\mathbf{U}_t + \beta_t\mathbf{M}_{t+1}\big) \\ &= \mathbf{z}_t + \beta_t\big(\mathbf{v}(\mathbf{x}_t) + \mathbf{U}_t\big) \\ &= \mathbf{z}_t + \beta_t\big(\mathbf{v}(\mathbf{z}_t) + \mathbf{e}_t\big),\end{aligned}$$

where

$$\mathbf{e}_t := \mathbf{v}(\mathbf{x}_t) - \mathbf{v}(\mathbf{z}_t) + \mathbf{U}_t.$$

By the $(1/\lambda)$-Lipschitz property of $\mathbf{v}$,

$$\|\mathbf{e}_t\| \leq \|\mathbf{v}(\mathbf{x}_t) - \mathbf{v}(\mathbf{z}_t)\| + \|\mathbf{U}_t\| \leq \Big(1 + \frac{1}{\lambda}\Big)\|\mathbf{U}_t\|.$$

Hence, using part (a),

$$\mathbb{E}\left[\|\mathbf{e}_t\|^2\right] \le \left(1+\frac{1}{\lambda}\right)^2 \mathbb{E}\left[\|\mathbf{U}_t\|^2\right] \le \left(1+\frac{1}{\lambda}\right)^2 \Gamma_3 \beta_t.$$

Since

$$\mathbf{z}_{t+1} - \mathbf{z}_t = \beta_t\big(\mathbf{v}(\mathbf{z}_t) + \mathbf{e}_t\big),$$

we have

$$\mathbf{v}(\mathbf{z}_t) = \frac{\mathbf{z}_{t+1} - \mathbf{z}_t - \beta_t \mathbf{e}_t}{\beta_t}.$$

Therefore

$$\begin{aligned}\|\mathbf{v}(\mathbf{z}_{t+1})\|^2 &= \|\mathbf{v}(\mathbf{z}_t)\|^2 + \|\mathbf{v}(\mathbf{z}_{t+1}) - \mathbf{v}(\mathbf{z}_t)\|^2 + 2\langle \mathbf{v}(\mathbf{z}_{t+1}) - \mathbf{v}(\mathbf{z}_t), \mathbf{v}(\mathbf{z}_t)\rangle \\ &= \|\mathbf{v}(\mathbf{z}_t)\|^2 + \|\mathbf{v}(\mathbf{z}_{t+1}) - \mathbf{v}(\mathbf{z}_t)\|^2 \\ &\quad + \frac{2}{\beta_t}\Big\langle \mathbf{v}(\mathbf{z}_{t+1}) - \mathbf{v}(\mathbf{z}_t), \mathbf{z}_{t+1} - \mathbf{z}_t - \beta_t \mathbf{e}_t \Big\rangle.\end{aligned}$$

Since $-\mathbf{v}$ is λ-co-coercive,

$$\langle \mathbf{z}_{t+1} - \mathbf{z}_t, \mathbf{v}(\mathbf{z}_{t+1}) - \mathbf{v}(\mathbf{z}_t)\rangle \le -\lambda \|\mathbf{v}(\mathbf{z}_{t+1}) - \mathbf{v}(\mathbf{z}_t)\|^2.$$

Substituting this above yields

$$\|\mathbf{v}(\mathbf{z}_{t+1})\|^2 \le \|\mathbf{v}(\mathbf{z}_t)\|^2 - \Big(\frac{2\lambda}{\beta_t} - 1\Big)\|\mathbf{v}(\mathbf{z}_{t+1}) - \mathbf{v}(\mathbf{z}_t)\|^2 - 2\langle \mathbf{v}(\mathbf{z}_{t+1}) - \mathbf{v}(\mathbf{z}_t), \mathbf{e}_t\rangle.$$

Now note that for any $c > 0$,

$$0 \le \Big\|\sqrt{c}\, u + \frac{1}{\sqrt{c}}\, w\Big\|^2 = c\|u\|^2 + 2\langle u, w\rangle + \frac{1}{c}\|w\|^2,$$

which implies

$$-c\|u\|^2 - 2\langle u, w\rangle \le \frac{1}{c}\|w\|^2.$$

Applying this with

$$c = \frac{2\lambda}{\beta_t} - 1, \qquad u = \mathbf{v}(\mathbf{z}_{t+1}) - \mathbf{v}(\mathbf{z}_t), \qquad w = \mathbf{e}_t,$$

we get

$$\|\mathbf{v}(\mathbf{z}_{t+1})\|^2 \le \|\mathbf{v}(\mathbf{z}_t)\|^2 + \frac{1}{c}\|\mathbf{e}_t\|^2.$$

Since $T_0 \ge 1/\lambda$, $\beta_t \le \lambda$ for all t, we have

$$c = \frac{2\lambda}{\beta_t} - 1 \ge \frac{\lambda}{\beta_t},$$

and hence

$$\|\mathbf{v}(\mathbf{z}_{t+1})\|^2 \le \|\mathbf{v}(\mathbf{z}_t)\|^2 + \frac{\beta_t}{\lambda}\|\mathbf{e}_t\|^2.$$

Taking expectation and using the bound on $\mathbb{E}[\|\mathbf{e}_t\|^2]$, we obtain

$$\mathbb{E}\left[\|\mathbf{v}(\mathbf{z}_{t+1})\|^2\right] \le \mathbb{E}\left[\|\mathbf{v}(\mathbf{z}_t)\|^2\right] + \frac{(1+\lambda^{-1})^2\Gamma_3}{\lambda}\beta_t^2.$$

Define

$$\Gamma_5 := \frac{(1+\lambda^{-1})^2\Gamma_3}{\lambda}.$$

Then

$$\mathbb{E}\left[\|\mathbf{v}(\mathbf{z}_{t+1})\|^2\right] \le \mathbb{E}\left[\|\mathbf{v}(\mathbf{z}_t)\|^2\right] + \Gamma_5 \beta_t^2.$$

Iterating this inequality from $j = i$ to $t-1$ gives

$$\mathbb{E}\left[\|\mathbf{v}(\mathbf{z}_i)\|^2\right] \ge \mathbb{E}\left[\|\mathbf{v}(\mathbf{z}_t)\|^2\right] - \Gamma_5 \sum_{j=i}^{t-1} \beta_j^2,$$

which proves part (c). □